

16) a) If A and B are 3×3 and $B = [b_1 \ b_2 \ b_3]$, then $AB = [Ab_1 + Ab_2 + Ab_3]$.

Solution: By defn. of matrix multiplication on Page 97,

$$AB = [Ab_1 \ Ab_2 \ Ab_3] \neq [Ab_1 + Ab_2 + Ab_3].$$

$\therefore$ The given statement is False.

b) The second row of AB is the second row of A multiplied on the right by B .

Solution: By the Row-Column Rule for computing AB on Page 98, for any arbitrary column j of AB , $(AB)_{2j} = a_{21}b_{1j} + a_{22}b_{2j} + a_{23}b_{3j} + \dots + a_{2n}b_{nj} + \dots$

$\therefore$ The 2nd row, j th column of AB is the multiplication of the 2nd row of A by the j th column of B on the right.

So, the 2nd row of AB is the second row of A multiplied on the right by B .

$\therefore$ The given statement is True.

c) $(AB)C = (AC)B$

Solution: $(AB)C = A(BC)$ (Theorem 2(a) on Page 99) $\neq A(BC)$ (Page 100, on the paragraph above Example 7) $= (AC)B$ (Theorem 2(a) on Page 99)

$\therefore (AB)C \neq (AC)B$. The given statement is False.

d) $(AB)^T = A^T B^T$

Solution: On Page 101 just below Theorem 3, it is mentioned that usually $(AB)^T \neq A^T B^T$. $\therefore$ The given statement is False.

e) The transpose of a sum of matrices equals the sum of their transposes.

Solution: From Theorem 3(b) on Page 101, the given statement is True.

17) If $A = \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix}$ and $AB = \begin{bmatrix} 1 & 2 & -1 \\ 6 & -9 & 3 \end{bmatrix}$, determine the first and second columns of B .

Solution: A is a 2×2 matrix. AB is a 2×3 matrix. From the defn. of matrix multiplication, B must have 2 rows and 3 columns.